

Advanced Lesson 7 - Exercises: Internal Form, Boolean Tests, Attributes

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MATH 7013 - Advanced Mathematical Modeling

1: Head and FullForm of E^x and x^E

```
Clear[x];  
{Head[ex], FullForm[ex]}  
  
{Head[xe], FullForm[xe]}
```

2: Head, FullForm and TreeForm of $x + y^2 \{u, v, w\}$

```
Clear[x, y, u, v, w];  
x + y2 {u, v, w}  
  
Head[x + y2 {u, v, w}]  
  
FullForm[x + y2 {u, v, w}]  
  
TreeForm[x + y2 {u, v, w}]
```

3: Is Pi an integer, a rational, a real?

```
Head[ $\pi$ ]  
  
NumericQ[ $\pi$ ]
```

```
NumberQ[ $\pi$ ]
```

```
Positive[ $\pi$ ]
```

4: Is "This is a string" a list or a string?

```
ListQ["This is a string"]
```

```
StringQ["This is a string"]
```

5: Is 3 a member of {1, 2, 3, 4, 5}?

```
MemberQ[{1, 2, 3, 4, 5}, 3]
```

6: A sign function built from Boolean tests

```
Clear[f, x];
f[x_] := +1 /; Positive[x];
f[0] = 0;
f[x_] := -1 /; Negative[x]
??f

{f[-2], f[0.023`], f[0], f[0.`], f[ $\pi$ ], f[3 + 4 i], f[a]}

Plot[f[x], {x, -2, 2}]
```

7: Attributes, and why $x = 2$ hurts Integrate but not Plot

```
Attributes[Plot]

Attributes[Integrate]

Clear[x];
x = 2;
Plot[x2, {x, 0, 1}]

Plot[ $\int \sin[x] \, dx$ , {x, 0, 1}]

Clear[x];
Plot[Evaluate[ $\int \sin[x] \, dx$ ], {x, 0, 1}]
```

8: Table and Integrate together

```
Attributes[Table]
Attributes[Integrate]

Clear[x, n];
x = 2;
n = 3;
Table[ $\int \sin[nx] dx$ , {n, 0, 10}]

Clear[x];
n = 3;
Table[ $\int \sin[nx] dx$ , {n, 0, 10}]
```

9: Locating subexpressions with Position

```
Clear[x, y, u, v];
x + y2 {u, v}
TreeForm[x + y2 {u, v}]
Position[x + y2 {u, v}, x + u y2]
Position[x + y2 {u, v}, u y2]
Position[x + y2 {u, v}, y2]
Position[x + y2 {u, v}, u]
Position[x + y2 {u, v}, y]

Clear[x, y, u, v];
TableForm[Table[TreeForm[(x + y2 {u, v})[[i]]], {i, 1, 2}]]
TableForm[Table[TreeForm[(x + y2 {u, v})[[i, j]]], {i, 1, 2}, {j, 1, 2}]]

Clear[x, y, u, v];
TreeForm[{x + u y2, x + v y2}]
{x + u y2, x + v y2}[[0]]
{x + u y2, x + v y2}[[1, 0]]
{x + u y2, x + v y2}[[1, 2, 0]]
{x + u y2, x + v y2}[[1, 2, 2, 0]]
```